Alireza Jafartash

Senior Back-End Software Engineer

Tehran, Iran | Open to Remote
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Summary

Back-end engineer with 4+ years of production experience, working across different technologies, programming languages, databases, paradigms, and products. Builds concurrency-safe financial systems, high-throughput APIs, and real-time data pipelines, and also builds front ends for full-stack products. M.Sc. candidate in Computer Engineering.

Skills

- **Primary:** Go (Golang), Python (Django), Java 21 (Spring Boot 3), TypeScript (NestJS, Next.js, React, Node.js), PostgreSQL (High Availability, Replication, Partitioning, Query Optimization), Redis, Apache Kafka, RabbitMQ, REST, Docker, Clean Architecture, SOLID, Concurrency Control
- **Familiar:** Kotlin, MongoDB, GraphQL, API Gateways & Rate Limiting, Kubernetes, Terraform, GCP, Linux, CI/CD, Prometheus, Grafana, Liquibase

Work Experience

Consultant and Lead Back-End Developer
Independent
2025–Present · Tehran

- Built an early-stage multi-currency trading SaaS for a few jewelry merchants as a full-stack NestJS and Next.js product, with integer minor-unit arithmetic, pessimistic row locks, and an append-only journal so concurrent orders never oversell stock, processing 10,000 daily transactions with zero balance drift.
- Built a full-stack multi-site kitchen inventory system with a single authoritative ledger for intake, transfers, waste, and usage across freezers, prep lines, and ambient stock for multiple sites, plus a notification system and monitoring and tracking across all sites.

Tools: Go, NestJS, PostgreSQL, Redis, Next.js, React, Bun

Software Engineer
LvtGames
Aug 2025–Mar 2026 · Istanbul, Türkiye · Remote · Full-time

- Built high-throughput transaction processing and multi-balance wallet services in Go 1.24 and Spring Boot 3 (Java 21) for a multi-tenant B2B2C entertainment platform.
- Provisioned GCP infrastructure with Terraform, implemented Bitbucket layer-cached CI/CD pipelines that cut deployment times by ~30%, and led cross-functional feature launches.

Tools: Go 1.24, Java 21, Spring Boot 3, PostgreSQL, Redis, GCP, Terraform, Liquibase

Software Engineer
SwapWallet
May 2025–Jul 2025 · Tehran · Full-time

- Resolved critical ledger concurrency race conditions and optimized the crypto exchange matching engine under peak load, reducing p99 execution latency.
- Automated administrative and member operations by designing an event-driven Telegram workflow bot integrated with core exchange services.

Tools: Kotlin, Spring Framework, Apache Kafka, PostgreSQL, Kubernetes

Software Engineer
Ultratrader
Feb 2024–Oct 2024 · Toronto, Canada · Remote · Full-time

- Designed a real-time ingestion pipeline that streams and normalizes financial market data across 30+ external broker APIs at ~1,000 events/sec sustained, with bursts to 300k events/sec.
- Built REST and GraphQL APIs alongside low-latency broker proxy gateways, and optimized analytical PostgreSQL queries to cut reporting latency by more than 90%.

Tools: NestJS, TypeScript, PostgreSQL, Redis, GraphQL, Docker

Software Engineer
Ravindex
May 2022–Dec 2023 · Tehran

- Ingested and processed stock-market tick data through high-throughput analytical pipelines backed by replicated PostgreSQL databases.
- Developed event-driven microservices using Django and RabbitMQ across 20+ automated data-fetching workers, instrumented with Prometheus and Grafana alerting.

Tools: Python, Django, PostgreSQL, RabbitMQ, Docker, Prometheus, Grafana

Research and Development Intern
Plotset
Aug 2021–Jul 2022 · California, US · Remote
- Researched AI-driven data visualization and architected a modular rendering framework integrating multiple charting engines in Node.js and D3.js.

Other Experience

Head Teaching Assistant — Advanced Programming
University of Tehran
2026–Present · Tehran
- Lead a team of 15+ TAs across problem design, code reviews, grading, and curriculum delivery for 200+ undergraduate students per semester.
- Designed and maintain the containerized Docker CI judge and automated assignment evaluation toolchain used across all terms for Java OOP, SOLID, and concurrency coursework.

Co-Founder & Lead Architect
Hambazi — Boardgame & Startup Community
Aug 2025–Present · Tehran
- Built a Telegram bot for user, event, and game management, backed by the community platform backend (NestJS and Next.js) that supports event scheduling, matchmaking, and member workflows for 1,000+ active members.

Education

M.Sc. in Computer Engineering — Algorithms and Computation
University of Tehran · Sep 2024–Present
Research focus: Distributed algorithms and fault-tolerant coordination.

B.Sc. in Computer Engineering
Babol Noshirvani University of Technology · Sep 2018–Aug 2024
Thesis: High-Throughput Distributed UUID Generation System — Engineered a collision-free 128-bit (16-byte) distributed ID generator in Go and RabbitMQ sustaining 10,000,000 IDs/sec per node with automated node coordination.

Projects

Promux
Autonomous multiplexer for agentic AI CLIs with zero-downtime account rotation.

Architected a proxy multiplexer that intercepts agentic CLI calls. Automatically detects HTTP 429/500 quota exhaustion, triggers account rotation with TTL cooling, and preserves running agent session context without interrupting long code generation loops.

Go, CLI, AI Systems

Manova / Orbit Platform
Shared foundation for client products: auth, limits, billing, and tenant isolation.

Shared micro-platform for client products, with RBAC, token-bucket rate limits, billing journals, and partitioned PostgreSQL for multi-tenant isolation across 5+ applications.

TypeScript, Go, PostgreSQL, Edge

Dorost
Open-source bilingual document composition package on official Typst registry.

Authored and published an open-source Typst package providing zero-dependency RTL/LTR mixed-direction layout, numeral conversion, and design system components with automated CI test suites.

Typst, Open Source, Design System

Awards & Certifications

2nd Place Golang Software Engineering Selection — Bale (2023)
Top 20 DevOps Boot Camp Selection — Digikala (2022)
Top 30 CyberSecurity Competition & Advanced Banking Security Workshop — Quera (2023)
Top 30 Software Engineering Bootcamp (Backend, APIs, DB Architecture) — Digikala (Sep 2022)
Top 20 DiffCode Algorithmic Contest — Quera (2019)
Certificate Machine Learning Specialization — DeepLearning.AI (Mar 2023)